

The Z-Transform and Its Properties

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Reference:

Sections 3.1 and 3.2 of

John G. Proakis and Dimitris G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th edition, 2007.

The Direct z-Transform

- ▶ Direct z-Transform:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

- ▶ Notation:

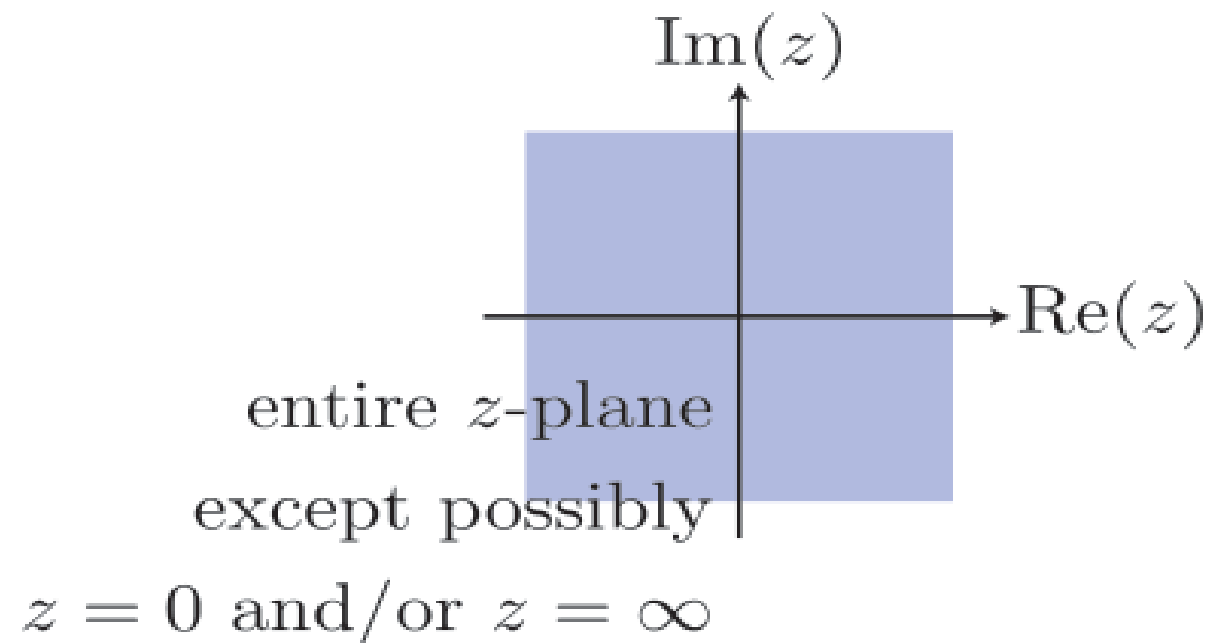
$$X(z) \equiv \mathcal{Z}\{x(n)\}$$

$$x(n) \xleftrightarrow{\mathcal{Z}} X(z)$$

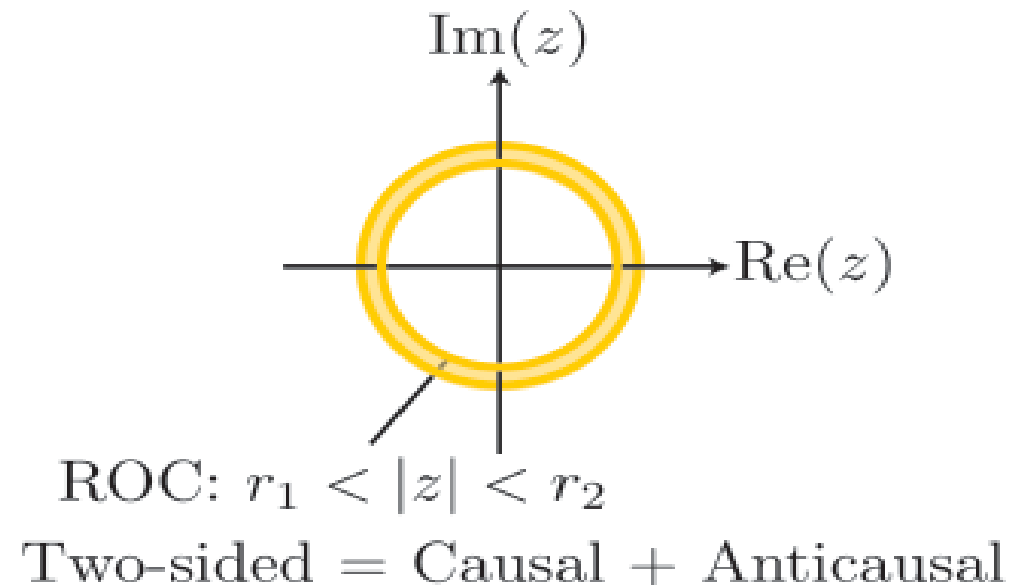
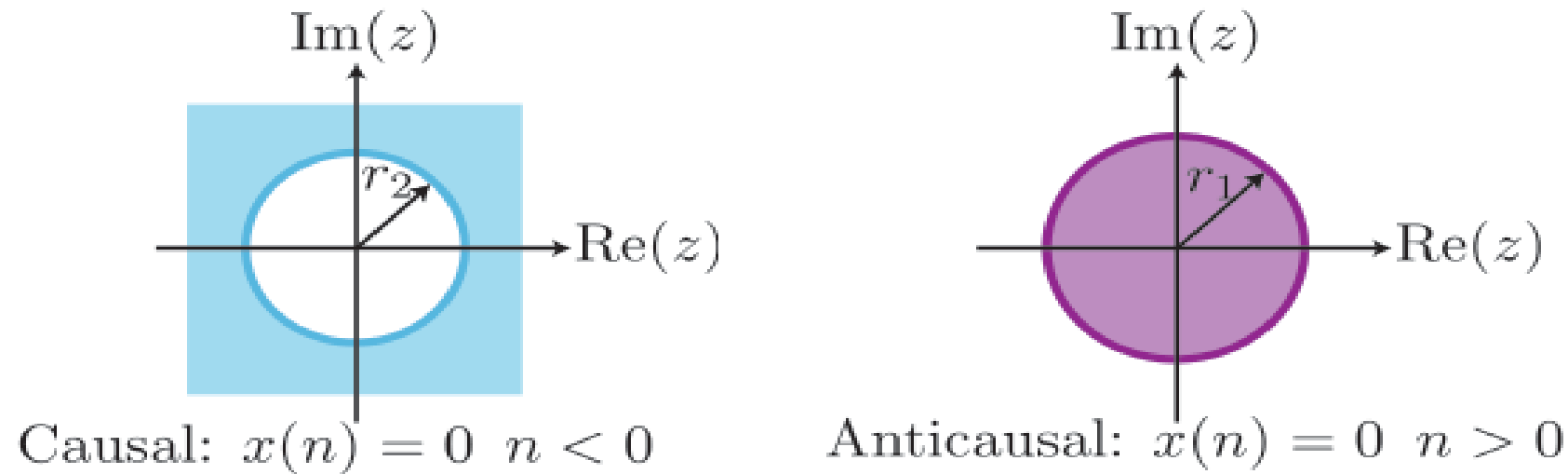
Region of Convergence

- ▶ the region of convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value
- ▶ The z-Transform is, therefore, uniquely characterized by:
 1. expression for $X(z)$
 2. ROC of $X(z)$

ROC Families: Finite Duration Signals



ROC Families: Infinite Duration Signals



z-Transform Properties

Property	Time Domain	z-Domain	ROC
Notation:	$x(n)$ $x_1(n)$ $x_2(n)$	$X(z)$ $X_1(z)$ $X_2(z)$	ROC: $r_2 < z < r_1$ ROC_1 ROC_2
Linearity:	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(z) + a_2 X_2(z)$	At least $\text{ROC}_1 \cap \text{ROC}_2$
Time shifting:	$x(n - k)$	$z^{-k} X(z)$	At least ROC, except $z = 0$ (if $k > 0$) and $z = \infty$ (if $k < 0$)
z-Scaling:	$a^n x(n)$	$X(a^{-1} z)$	$ a r_2 < z < a r_1$
Time reversal	$x(-n)$	$X(z^{-1})$	$\frac{1}{r_1} < z < \frac{1}{r_2}$
Conjugation:	$x^*(n)$	$X^*(z^*)$	ROC
z-Differentiation:	$n x(n)$	$-z \frac{dX(z)}{dz}$	$r_2 < z < r_1$
Convolution:	$x_1(n) * x_2(n)$	$X_1(z) X_2(z)$	At least $\text{ROC}_1 \cap \text{ROC}_2$

among others ...

Linearity Property

$$x_1(n) \xleftrightarrow{z} X_1(z), \text{ROC}_1$$

$$x_2(n) \xleftrightarrow{z} X_2(z), \text{ROC}_2$$

$$a_1x_1(n) + a_2x_2(n) \xleftrightarrow{z} a_1X_1(z) + a_2X_2(z),$$

At least $\text{ROC}_1 \cap \text{ROC}_2$

Linearity Property

Let $x(n) = a_1x_1(n) + a_2x_2(n)$

$$\begin{aligned}X(z) &= \sum_{n=-\infty}^{\infty} x(n)z^{-n} \\&= \sum_{n=-\infty}^{\infty} (a_1x_1(n) + a_2x_2(n))z^{-n} \\&= a_1 \sum_{n=-\infty}^{\infty} x_1(n)z^{-n} + a_2 \sum_{n=-\infty}^{\infty} x_2(n)z^{-n} \\&= a_1X_1(z) + a_2X_2(z)\end{aligned}$$

ROC: At least $\text{ROC}_1 \cap \text{ROC}_2$

Time Shifting Property

$$x(n) \xleftrightarrow{z} X(z), \quad \text{ROC}$$

$$x(n - k) \xleftrightarrow{z} z^{-k} X(z), \quad \begin{array}{l} \text{At least ROC} \\ \text{except } z = 0 \ (k > 0) \text{ or} \\ z = \infty \ (k < 0) \end{array}$$

Time Shifting Property

Let $y(n) = x(n - k)$.

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} (x(n - k)) z^{-n} \end{aligned}$$

Let $m = n - k$

$$\begin{aligned} &= \sum_{m=-\infty}^{\infty} x(m) z^{-(m+k)} = \sum_{m=-\infty}^{\infty} x(m) z^{-m} z^{-k} \\ &= z^{-k} \left[\sum_{m=-\infty}^{\infty} x(m) z^{-m} \right] = z^{-k} X(z) \end{aligned}$$

Time Shifting Property: ROC

$$x(n) = \delta(n) \xleftrightarrow{z} X(z) = 1, \text{ROC: } \underline{\text{entire } z\text{-plane}}$$

- Example: For $k = -1$

$$y(n) = x(n - (-1)) = x(n + 1) = \delta(n + 1)$$

$$y(n) = \delta(n + 1) \xleftrightarrow{z} Y(z) = z, \text{ROC: } \underline{\text{entire } z\text{-plane}} \\ \text{except } z = \infty$$

- Example: For $k = 1$

$$y(n) = x(n - (1)) = x(n - 1) = \delta(n - 1)$$

$$y(n) = \delta(n - 1) \xleftrightarrow{z} Y(z) = z^{-1}, \text{ROC: } \underline{\text{entire } z\text{-plane}} \\ \text{except } z = 0$$

Scaling in the z-Domain

$$x(n) \xleftrightarrow{z} X(z), \quad \text{ROC: } r_1 < |z| < r_2$$

$$a^n x(n) \xleftrightarrow{z} X(a^{-1}z), \quad \text{ROC: } |a|r_1 < |z| < |a|r_2$$

Scaling in the z-Domain

Let $y(n) = a^n x(n)$

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} a^n x(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} x(n) (a^{-1} z)^{-n} \\ &= X(a^{-1} z) \end{aligned}$$

$$\text{ROC: } r_1 < |a^{-1} z| = \frac{|z|}{|a|} < r_2 \quad \equiv \quad |a| r_1 < |z| < |a| r_2$$

z-Transform Properties

Property	Time Domain	z-Domain	ROC
Notation:	$x(n)$ $x_1(n)$ $x_2(n)$	$X(z)$ $X_1(z)$ $X_2(z)$	ROC: $r_2 < z < r_1$ ROC_1 ROC_2
Linearity:	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(z) + a_2 X_2(z)$	At least $\text{ROC}_1 \cap \text{ROC}_2$
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z-Differentiation:	$n x(n)$	$-z \frac{dX(z)}{dz}$	$r_2 < z < r_1$
Convolution:	$x_1(n) * x_2(n)$	$X_1(z) X_2(z)$	At least $\text{ROC}_1 \cap \text{ROC}_2$

among others ...

Convolution Property

$$x(n) = x_1(n) * x_2(n) \iff X(z) = X_1(z) \cdot X_2(z)$$

Convolution using the z-Transform

Basic Steps:

1. Compute z-Transform of each of the signals to convolve (time domain $\rightarrow$ z-domain):

$$X_1(z) = \mathcal{Z}\{x_1(n)\}$$

$$X_2(z) = \mathcal{Z}\{x_2(n)\}$$

2. Multiply the two z-Transforms (in z-domain):

$$X(z) = X_1(z)X_2(z)$$

3. Find the inverse z-Transform of the product (z-domain $\rightarrow$ time domain):

$$x(n) = \mathcal{Z}^{-1}\{X(z)\}$$

Common Transform Pairs

	Signal, $x(n)$	z-Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1-z^{-1}}$	$ z > 1$
3	$a^n u(n)$	$\frac{1}{1-az^{-1}}$	$ z > a $
4	$na^n u(n)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
5	$-a^n u(-n-1)$	$\frac{1}{1-az^{-1}}$	$ z < a $
6	$-na^n u(-n-1)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
7	$\cos(\omega_0 n) u(n)$	$\frac{1-z^{-1} \cos \omega_0}{1-2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
8	$\sin(\omega_0 n) u(n)$	$\frac{z^{-1} \sin \omega_0}{1-2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
9	$a^n \cos(\omega_0 n) u(n)$	$\frac{1-az^{-1} \cos \omega_0}{1-2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
10	$a^n \sin(\omega_0 n) u(n)$	$\frac{1-az^{-1} \sin \omega_0}{1-2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

Common Transform Pairs

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10	$a^n \sin(\omega_0 n) u(n)$	$\frac{1-az^{-1} \sin \omega_0}{1-2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

Common Transform Pairs

- ▶ z-Transform expressions that are a fraction of polynomials in z^{-1} (or z) are called **rational**.
- ▶ z-Transforms that are **rational** represent an important class of signals and systems.



Rational Z-Transforms and Its Inverse

Reference

Reference:

Sections 3.3 and 3.4 of

John G. Proakis and Dimitris G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th edition, 2007.

Rational z-Transforms

Why Rational?

- ▶ $X(z)$ is a rational function iff it can be represented as the ratio of two polynomials in z^{-1} (or z):

$$X(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}}$$

- ▶ For LTI systems that are represented by **LCCDEs**, the z-Transform of the unit sample response $h(n)$, denoted $H(z) = \mathcal{Z}\{h(n)\}$, is **rational**

Poles and Zeros

- ▶ zeros of $X(z)$: values of z for which $X(z) = 0$
- ▶ poles of $X(z)$: values of z for which $X(z) = \infty$

Poles and Zeros of the Rational z-Transform

Let $a_0, b_0 \neq 0$:

$$\begin{aligned}
 X(z) = \frac{B(z)}{A(z)} &= \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}} \\
 &= \left(\frac{b_0 z^{-M}}{a_0 z^{-N}} \right) \frac{z^M + (b_1/b_0)z^{M-1} + \dots + b_M/b_0}{z^N + (a_1/a_0)z^{N-1} + \dots + a_N/a_0} \\
 &= \frac{b_0}{a_0} z^{-M+N} \frac{(z - z_1)(z - z_2) \dots (z - z_M)}{(z - p_1)(z - p_2) \dots (z - p_N)} \\
 &= G z^{N-M} \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)}
 \end{aligned}$$

Poles and Zeros of the Rational z-Transform

$$X(z) = G z^{N-M} \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)} \quad \text{where } G \equiv \frac{b_0}{a_0}$$

- ▶ $X(z)$ has M **finite zeros** at $z = z_1, z_2, \dots, z_M$
- ▶ $X(z)$ has N **finite poles** at $z = p_1, p_2, \dots, p_N$
- ▶ For $N - M \neq 0$
 - ▶ if $N - M > 0$, there are $|N - M|$ **zeros** at **origin**, $z = 0$
 - ▶ if $N - M < 0$, there are $|N - M|$ **poles** at **origin**, $z = 0$

Total number of zeros = Total number of poles

Poles and Zeros of the Rational z-Transform

Example:

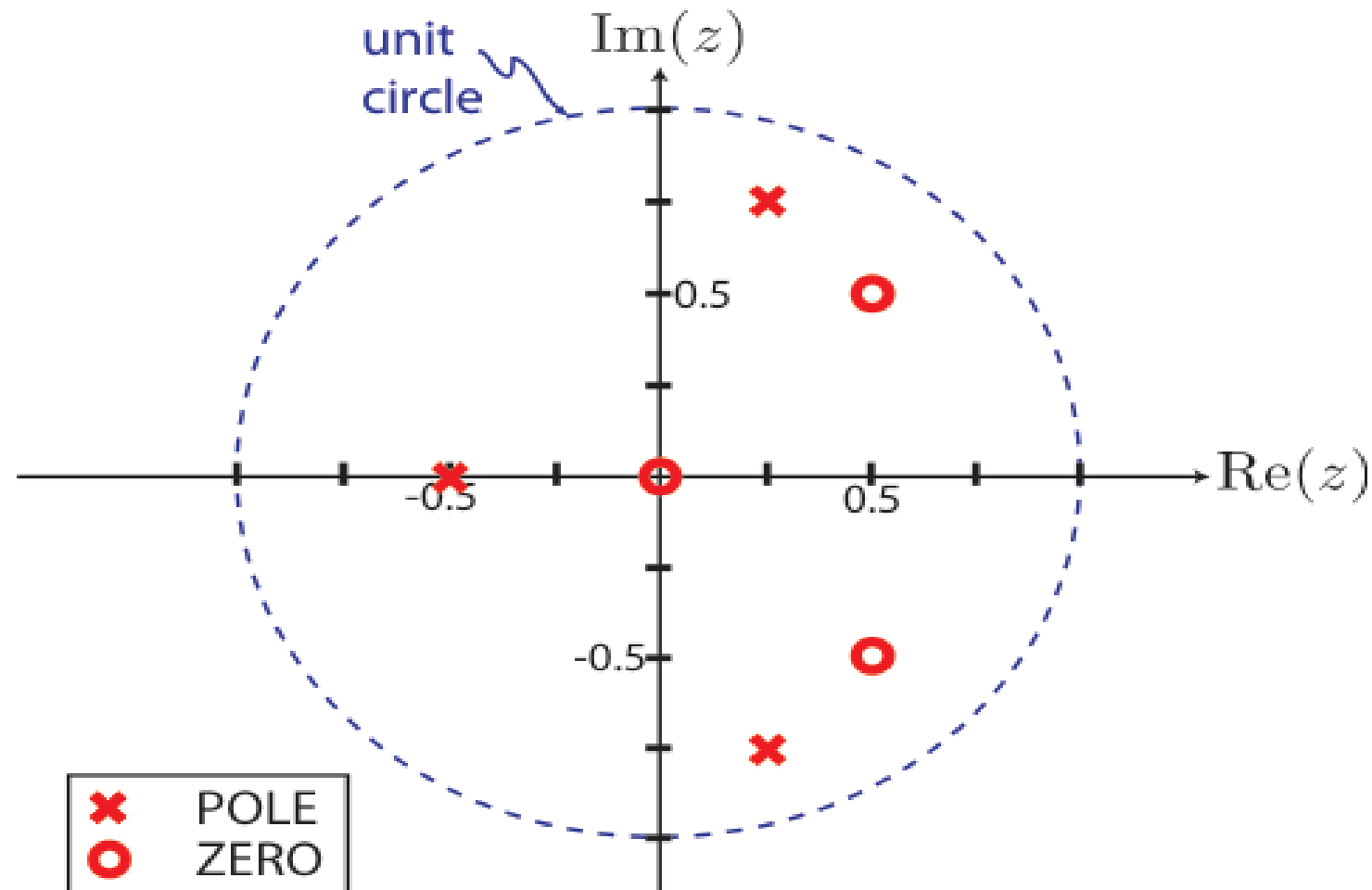
$$\begin{aligned} X(z) &= z \frac{2z^2 - 2z + 1}{16z^3 + 6z + 5} \\ &= (z - 0) \frac{(z - (\frac{1}{2} + j\frac{1}{2}))(z - (\frac{1}{2} - j\frac{1}{2}))}{(z - (\frac{1}{4} + j\frac{3}{4}))(z - (\frac{1}{4} - j\frac{3}{4}))(z - (-\frac{1}{2}))} \end{aligned}$$

$$\text{poles: } z = \frac{1}{4} \pm j\frac{3}{4}, -\frac{1}{2}$$

$$\text{zeros: } z = 0, \frac{1}{2} \pm j\frac{1}{2}$$

Pole-Zero Plot

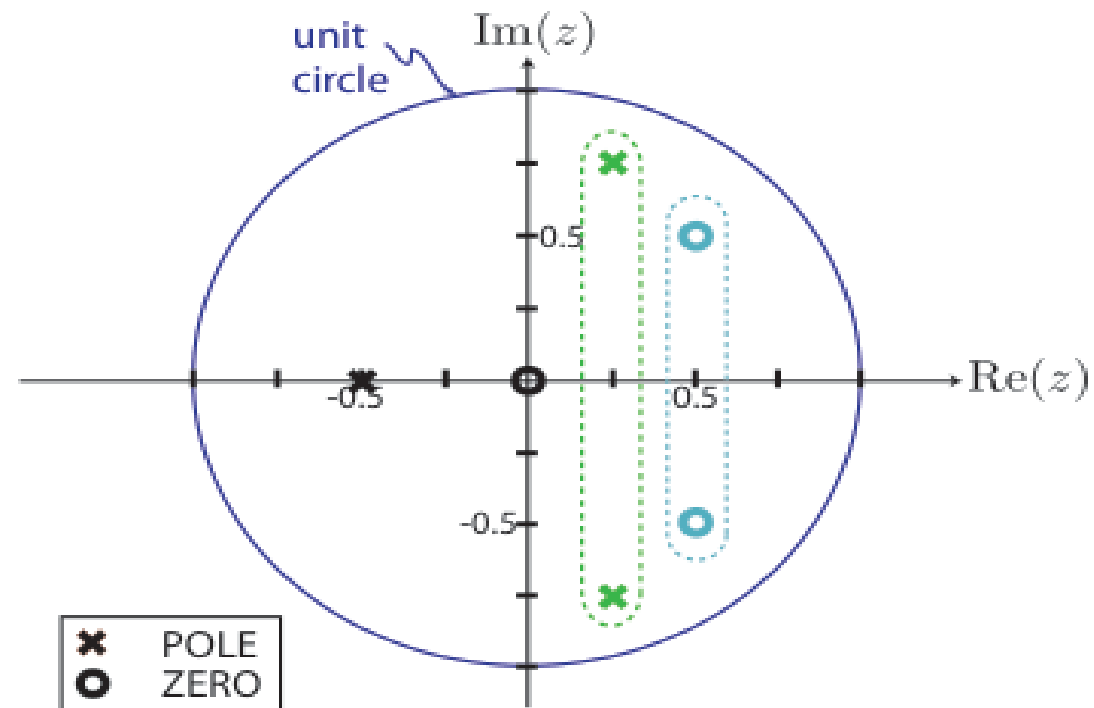
Example: poles: $z = \frac{1}{4} \pm j\frac{3}{4}, -\frac{1}{2}$, zeros: $z = 0, \frac{1}{2} \pm j\frac{1}{2}$



Pole-Zero Plot and Conjugate Pairs

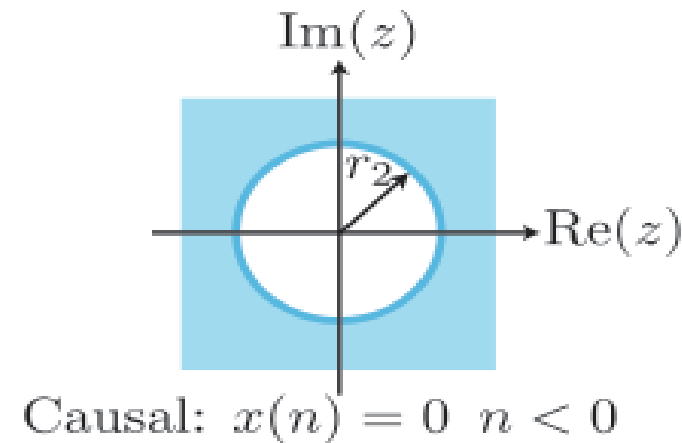
- ▶ For **real** time-domain signals, the coefficients of $X(z)$ are necessarily **real**
 - ▶ complex poles and zeros must occur in **conjugate pairs**
 - ▶ note: real poles and zeros do not have to be paired up

$$X(z) = z \frac{2z^2 - 2z + 1}{16z^3 + 6z + 5} \Rightarrow$$

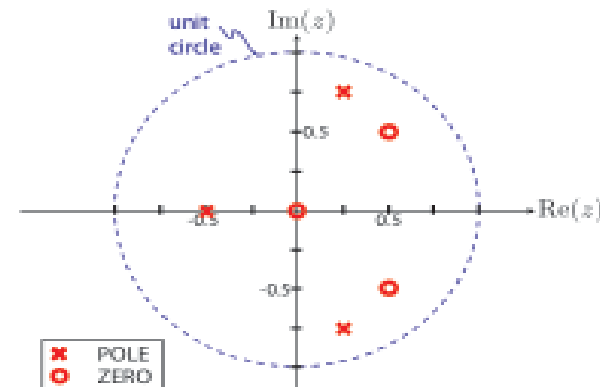


Pole-Zero Plot and the ROC

- Recall, for causal signals, the ROC will be the outer region of a disk

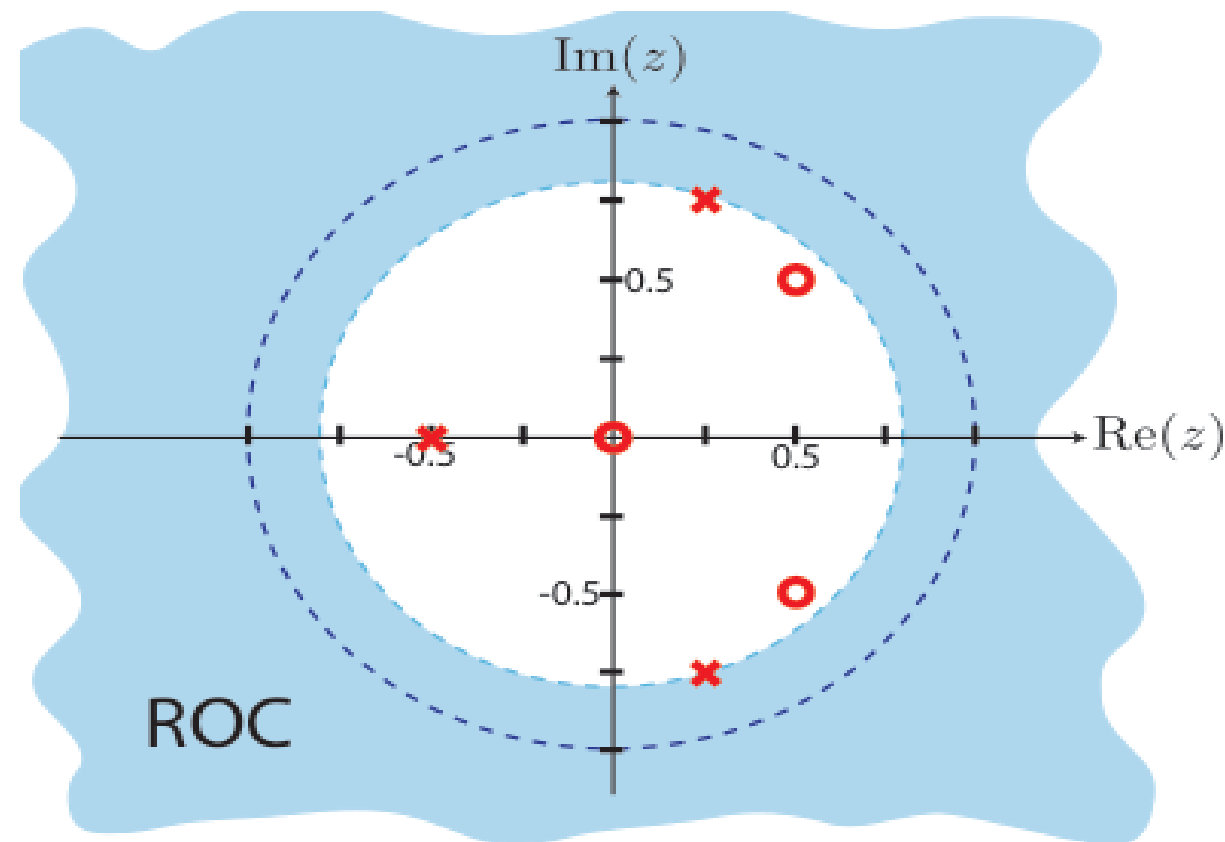


- ROC cannot necessarily include poles ($\because X(p_k) = \infty$)



Pole-Zero Plot and the ROC

- Therefore, for a **causal** signal the ROC is the **smallest** (origin-centered) circle encompassing all the poles.



Causality and Stability

- Recall,

$$\begin{array}{c} \text{LTI system is} \\ \text{stable} \end{array} \iff \sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

- Moreover,

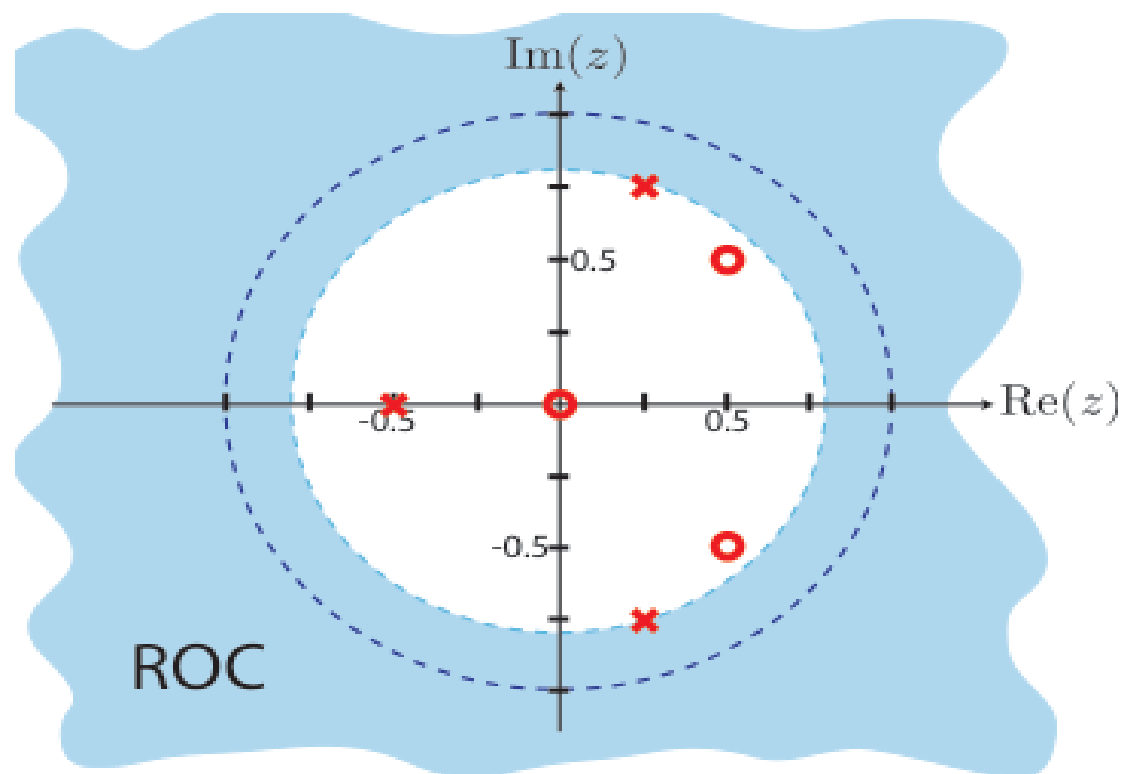
$$\begin{aligned} |H(z)| &= \left| \sum_{n=-\infty}^{\infty} h(n)z^{-n} \right| \leq \sum_{n=-\infty}^{\infty} |h(n)z^{-n}| \\ &= \sum_{n=-\infty}^{\infty} |h(n)| \quad \text{for } |z| = 1 \end{aligned}$$

- It can be shown:

$$\begin{array}{c} \text{LTI system is} \\ \text{stable} \end{array} \iff \sum_{n=-\infty}^{\infty} |h(n)| < \infty \iff \begin{array}{c} \text{ROC of } H(z) \text{ contains} \\ \text{unit circle} \end{array}$$

Pole-Zero Plot, Causality and Stability

- For stable systems, the ROC will include the unit circle.



- For stability of a **causal** system, the poles must lie **inside** the unit circle.

The System Function

$$h(n) \xleftrightarrow{z} H(z)$$

$$\text{time-domain} \xleftrightarrow{z} \text{z-domain}$$

$$\text{impulse response} \xleftrightarrow{z} \text{system function}$$

$$y(n) = x(n) * h(n) \xleftrightarrow{z} Y(z) = X(z) \cdot H(z)$$

Therefore,

$$H(z) = \frac{Y(z)}{X(z)}$$

The System Function of LCCDEs

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$$\mathcal{Z}\{y(n)\} = \mathcal{Z}\left\{- \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)\right\}$$

$$\mathcal{Z}\{y(n)\} = - \sum_{k=1}^N a_k \mathcal{Z}\{y(n-k)\} + \sum_{k=0}^M b_k \mathcal{Z}\{x(n-k)\}$$

$$Y(z) = - \sum_{k=1}^N a_k z^{-k} Y(z) + \sum_{k=0}^M b_k z^{-k} X(z)$$

The System Function of LCCDEs

$$\begin{aligned}
 Y(z) + \sum_{k=1}^N a_k z^{-k} Y(z) &= \sum_{k=0}^M b_k z^{-k} X(z) \\
 Y(z) \left[1 + \sum_{k=1}^N a_k z^{-k} \right] &= X(z) \sum_{k=0}^M b_k z^{-k} \\
 H(z) = \frac{Y(z)}{X(z)} &= \frac{\sum_{k=0}^M b_k z^{-k}}{\left[1 + \sum_{k=1}^N a_k z^{-k} \right]} \\
 \text{LCCDE} &\longleftrightarrow \text{Rational System Function}
 \end{aligned}$$

Many signals of practical interest have a rational z-Transform.

Inversion of the z-Transform

Inversion of the z-Transform

Three popular methods:

1. Contour integration:

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

2. Expansion into a **power series** in z or z^{-1} :

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{-k}$$

and obtaining $x(k)$ for all k by inspection.

3. Partial-fraction expansion and **table look-up**.

Expansion into Power Series

Example:

$$\begin{aligned} X(z) &= \log(1 + az^{-1}), \quad |z| > |a| \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n z^{-n}}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n}{n} z^{-n} \end{aligned}$$

By inspection:

$$x(n) = \begin{cases} \frac{(-1)^{n+1} a^n}{n} & n \geq 1 \\ 0 & n \leq 0 \end{cases}$$

Partial-Fraction Expansion

1. Find the distinct poles of $X(z)$: $p_1, p_2, \dots, p_K$ and their corresponding multiplicities $m_1, m_2, \dots, m_K$.
2. The partial-fraction expansion is of the form:

$$z^{-R}X(z) = \sum_{k=1}^K \left(\frac{A_{1k}}{z - p_k} + \frac{A_{2k}}{(z - p_k)^2} + \dots + \frac{A_{m_k}}{(z - p_k)^{m_k}} \right)$$

where p_k is an m_k th order pole (i.e., has multiplicity m_k) and R is selected to make $z^{-R}X(z)$ a strictly proper rational function.

3. Use an appropriate approach to compute $\{A_{ik}\}$

Partial-Fraction Expansion

Example: Find $x(n)$ given poles of $X(z)$ at $p_1 = -2$ and a double pole at $p_2 = p_3 = 1$; specifically,

$$\begin{aligned} X(z) &= \frac{1}{(1 + 2z^{-1})(1 - z^{-1})^2} \\ \frac{X(z)}{z} &= \frac{z^2}{(z + 2)(z - 1)^2} \\ \frac{z^2}{(z + 2)(z - 1)^2} &= \frac{A_1}{z + 2} + \frac{A_2}{z - 1} + \frac{A_3}{(z - 1)^2} \end{aligned}$$

Note: we need a strictly proper rational function.

DO NOT FORGET TO MULTIPLY BY z IN THE END.

Partial-Fraction Expansion

$$\begin{aligned}\frac{z^2(z+2)}{(z+2)(z-1)^2} &= \frac{A_1(z+2)}{z+2} + \frac{A_2(z+2)}{z-1} + \frac{A_3(z+2)}{(z-1)^2} \\ \frac{z^2}{(z-1)^2} &= A_1 + \frac{A_2(z+2)}{z-1} + \frac{A_3(z+2)}{(z-1)^2} \Big|_{z=-2} \\ A_1 &= \frac{4}{9}\end{aligned}$$

Partial-Fraction Expansion

$$\begin{aligned}\frac{z^2(z-1)^2}{(z+2)(z-1)^2} &= \frac{A_1(z-1)^2}{z+2} + \frac{A_2(z-1)^2}{z-1} + \frac{A_3(z-1)^2}{(z-1)^2} \\ \frac{z^2}{(z+2)} &= \frac{A_1(z-1)^2}{z+2} + A_2(z-1) + A_3 \Big|_{z=1} \\ A_3 &= \frac{1}{3}\end{aligned}$$

Partial-Fraction Expansion

$$\begin{aligned}\frac{z^2(z-1)^2}{(z+2)(z-1)^2} &= \frac{A_1(z-1)^2}{z+2} + \frac{A_2(z-1)^2}{z-1} + \frac{A_3(z-1)^2}{(z-1)^2} \\ \frac{z^2}{(z+2)} &= \frac{A_1(z-1)^2}{z+2} + A_2(z-1) + A_3 \\ \frac{d}{dz} \left[\frac{z^2}{(z+2)} \right] &= \frac{d}{dz} \left[\frac{A_1(z-1)^2}{z+2} + A_2(z-1) + A_3 \right] \Big|_{z=1} \\ A_2 &= \frac{5}{9}\end{aligned}$$

Partial-Fraction Expansion

Therefore, **assuming causality**, and using the following pairs:

$$\begin{aligned}a^n u(n) &\stackrel{z}{\longleftrightarrow} \frac{1}{1 - az^{-1}} \\na^n u(n) &\stackrel{z}{\longleftrightarrow} \frac{az^{-1}}{(1 - az^{-1})^2}\end{aligned}$$

$$\begin{aligned}X(z) &= \frac{4}{9} \frac{1}{1 + 2z^{-1}} + \frac{5}{9} \frac{1}{1 - z^{-1}} + \frac{1}{3} \frac{z^{-1}}{(1 - z^{-1})^2} \\x(n) &= \frac{4}{9} (-2)^n u(n) + \frac{5}{9} u(n) + \frac{1}{3} nu(n) \\&= \left[\frac{(-2)^{n+2}}{9} + \frac{5}{9} + \frac{n}{3} \right] u(n)\end{aligned}$$

